

PIYUSH KHOT

Full-Stack & Creative Developer

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TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C, C++, HTML5, CSS3

Frontend & Creative: React.js, Three.js, WebGL, GSAP, Anime.js, Tailwind CSS, Canvas API

Backend & Databases: Node.js, Express.js, MongoDB, REST APIs, SQLite

Tools & Systems: Git, GitHub, Vite, Postman, Linux/Bash, VS Code

FEATURED PROJECTS

1. Interactive 3D Solar System Explorer

Three.js, WebGL, JavaScript, Canvas API

- Built an interactive 3D planetary simulation featuring realistic orbits, dynamic lighting, and texture maps.
- Implemented smooth camera controls and interactive raycasting to inspect detailed planetary data on click.
- Optimized rendering performance to sustain a constant 60 FPS across desktop and mobile browsers.

2. TaskFlow - Full-Stack Task & Habit Tracker

React.js, Node.js, Express.js, MongoDB, JWT

- Developed a responsive CRUD web app with JWT authentication, password hashing, and user sessions.
- Engineered RESTful API routes supporting priority filters, deadline alerts, category grouping, and search.
- Crafted a clean, accessible UI workflow using React Hooks and modular CSS components.

3. Automated Price Tracker & Alert Bot

Python, BeautifulSoup4, Requests, SQLite, SMTPlib

- Built a scheduled web scraper monitoring e-commerce product price changes across multiple vendors.
- Stored historical pricing trends in SQLite and triggered automated email alerts when prices dropped below limits.
- Implemented robust error-handling, custom User-Agents, and exponential backoff against rate limiting.

4. Smart CLI File Organizer & System Monitor

Python (OS, Shutil, Psutil, Argparse), Bash

- Created a cross-platform CLI tool that automatically categorizes cluttered files by MIME type and creation date.
- Added real-time system resource tracking (CPU, RAM, disk) with automatic logging to CSV reports.
- Included CLI options like --dry-run, --undo, and interactive terminal indicators.

5. Memory Allocation & Cache Emulator

C++, STL, Valgrind, Make

- Simulated operating system memory allocation strategies (First-Fit, Best-Fit, Worst-Fit).
- Implemented LRU (Least Recently Used) and FIFO cache replacement policies to benchmark hit/miss performance.
- Verified leak-free memory management with Valgrind and strict compiler checks.

EXPERIENCE & MILESTONES

Creative & Full-Stack Development

2025 - Present

Foundational Software Engineering & Systems

2023 - 2024

EDUCATION

Bachelor of Technology / Computer Science

Completed